



NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2
Preliminary Examination

NAME

--

SUBJECT
CLASS

2ma2

REGISTRATION
NUMBER

--	--	--	--	--	--	--

H2 MATHEMATICS

9758 / 01

28 August 2019

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)
Writing Paper (1 sheet)

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number in the boxes above.
Please write clearly and use capital letters.

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the question paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved graphing calculator is expected, where appropriate.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.
Up to 2 marks may be deducted for improper presentation.

The number of marks is given in the brackets [] at the end of each question or part question.

Question Number	Marks Possible	Marks Obtained
1	6	
2	6	
3	7	
4	7	
5	8	
6(a)	3	
6(b)	6	
7	10	
8	10	
9	12	
10	12	
11	13	
Presentation Deduction		-1 / -2
TOTAL	100	

This document consists of **29** printed pages and **3** blank pages.

[Turn over

- 1 The complex number z has modulus r and argument θ , where $0 < r < 1$ and $\frac{\pi}{2} < \theta < \pi$. The complex number w is such that $\left|\frac{w}{z}\right| = 1$ and $\arg(w) + \arg(z) = \pi$.

Let the points A , B , C , D and E represent the complex numbers z , w , $z - w$, z^* and zz^* respectively, where z^* denotes the conjugate of z .

- (i) On a single Argand diagram, illustrate these five points. [5]
- (ii) Identify the shape of the quadrilateral $ACDE$. [1]

STAPLE HERE

STAPLE HERE

2 Do not use a calculator in answering this question.

Solve the inequality $\frac{2x^2 + 5x + 9}{x - 7} \leq x - 4$.

[4]

Hence, solve the inequality $\frac{2+5e^x+9e^{2x}}{7e^x-1} \geq 4e^x-1$.

[2]

STAPLE HERE

STAPLE HERE

[Turn over

- 3** Two sightseeing points A and B are located offshore. A cruise ship S travels along a straight path passing through A and B . A lighthouse is located at the shore. If the lighthouse's position is taken as the origin O , the position vectors of A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively.

- (i) If $AS : SB = \lambda : (1 - \lambda)$, find the position vector of S , giving your answer in terms of λ , $\mathbf{a}$ and $\mathbf{b}$. [1]

When the cruise ship is closest to the lighthouse,

- (ii) show that $\lambda = \frac{\mathbf{a} \cdot (\mathbf{a} - \mathbf{b})}{(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a})}$, [2]

- (iii) find its position vector in terms of $\mathbf{a}$ and $\mathbf{b}$, if sightseeing point A is three distance units away from the light house, sightseeing point B is two distance units away from the lighthouse and the angle between $\mathbf{a}$ and $\mathbf{b}$ is 60° . [4]

STAPLE HERE

STAPLE HERE

[Turn over

4 (i) Show that $\frac{px^2 + (4p - q)x + (4p + q)}{(1 - x)(2 + x)^2} = \frac{p}{1 - x} + \frac{q}{(2 + x)^2}.$ [1]

(ii) Find the values of p , q and r such that the series expansion of

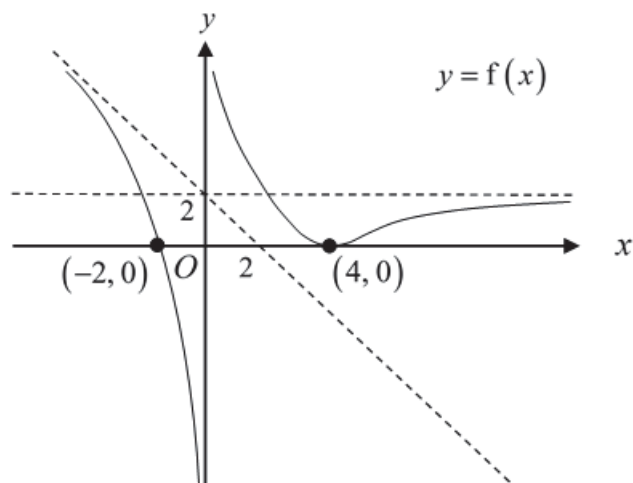
$$\frac{px^2 + (4p - q)x + (4p + q)}{(1 - x)(2 + x)^2} + r \ln(1 - 3x)$$

up to and including the term in x^2 is $11 - 3x + x^2.$ [4]

(iii) Find the range of the values of x for which the expansion is valid.

[2]

5



The diagram shows a sketch of the curve $y = f(x)$. The curve cuts the x -axis at $(-2, 0)$ and it has a stationary point at $(4, 0)$. The equations of the asymptotes are $x = 0$, $y = 2$ and $y = 2 - x$.

Sketch, in separate diagrams, the graphs of

(i) $y = \frac{1}{f(x)}$, [3]

(ii) $y = f'(x)$, [3]

stating the equations of any asymptotes and coordinates of axial intercepts.

In the curve $y = f(ax + b)$, there exists an oblique asymptote $y = 4 - 2x$, find the values of a and b . [2]

[Turn over

- 6 (a) Relative to the origin O , a point A has position vector $6\mathbf{i} + t\mathbf{j} + 10\mathbf{k}$, where t is a real and negative constant. Given that the direction cosine of A with respect to the x -axis is $\frac{3}{\sqrt{35}}$, find the value of t . Deduce the angle between $\overrightarrow{OA}$ and the yz -plane. [3]

- (b) The planes p_1 and p_2 have equations $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 8$ and $\mathbf{r} \cdot \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} = k$ respectively, where k is a real constant. It is given that the plane p is equidistant from p_1 and p_2 .

- (i) Find, in terms of k , a vector equation of p in the scalar product form. [3]

STAPLE HERE

STAPLE HERE

[Turn over

- (ii) Given $k = 13$, find the coordinates of the point on line l with equation

$$\frac{x-1}{3} = \frac{2y-1}{18} = \frac{3-z}{1}$$

that is equidistant from p_1 and p_2 .

[3]

7 Let $f(x) = \frac{x^2 + 3x + 4 - a}{x + a}$, where $0 < a < \frac{3}{2}$.

Sketch, in the separate diagrams, the graphs of

(i) $y = f(x)$, [3]

(ii) $y = |f(x)|$, [3]

stating the equations of the asymptotes and coordinates of the axial intercepts, in terms of a .

STAPLE HERE

STAPLE HERE

[Turn over

- (iii) It is given that $|f(x)| = m$ has exactly two distinct roots. Find the range of values of m in terms of a . [4]

STAPLE HERE

STAPLE HERE

[Turn over

- 8 A curve C has parametric equations

$$x = t^2 - 4t, \quad y = 3t^2 - t^3, \quad t \leq 0.$$

- (i) Sketch C .

[1]

A point P on C has parameter p .

- (ii) Show that the tangent at P has equation

$$y = -\frac{3}{2}px + \frac{1}{2}p^3 - 3p^2.$$

[3]

The tangent at P makes an angle of 45° with the x -axis.

(iii) Find the exact coordinates of P .

[2]

(iv) Find the area of the region bounded by the curve C , its tangent at P and the x -axis.

[4]

STAPLE HERE

STAPLE HERE

[Turn over

9 Functions f and g are defined by

$$f : x \mapsto 2|x - p| + 1 \text{ for } x \in \mathbb{R}, p > 1,$$

$$g : x \mapsto x(x - q) \text{ for } x \in \mathbb{R}, q < 0.$$

- (i) Explain why f does not have an inverse. [2]

The domain of f is now restricted to $x \leq k$.

- (ii) Write down the largest value of k for which the function f^{-1} exists. Hence find $f^{-1}(x)$ and state the domain of f^{-1} . [4]

- (iii) Sketch on the same diagram the graphs of $y=f(x)$ and $y=f^{-1}(x)$, giving the coordinates of the axial intercepts. Hence solve $f(x)=f^{-1}(x)$. [4]

[Turn over

(iv) Find the range of gf^{-1} .

[2]

- 10** A construction company, which has just won a contract to build foundations at two different sites I and II, wants to buy a new borewell drilling machine. Two brands, MTZ and STK, are considered and tested. To test a machine, it is used to drill a hole for ten hours.

The following data were recorded.

MTZ: The depth drilled is 4.5 m in the first hour, and for each subsequent hour, the depth drilled decreases by a positive constant $r\%$.

STK: The depth drilled is 4.7 m in the first hour, and for each subsequent hour, the depth drilled decreases by 0.1 m.

- (i)** Show that the total depth drilled by an MTZ borewell drilling machine is

$$\frac{450}{r} \left[1 - \left(1 - \frac{r}{100} \right)^{10} \right] \text{ m.} \quad [2]$$

- (ii)** Find the value of r if the total depth drilled by both machines are the same. [2]

STAPLE HERE

STAPLE HERE

[Turn over

It is given that the machines perform the same way as recorded at the test site.

- (iii) The company chooses a MTZ borewell drilling machine to build a foundation at Site I. Assuming $r=1$, to maximise the efficiency of the machine, the company will stop drilling if the depth to be drilled in the next hour is less than 4.0 m at this site. How many complete hours should the machine be operated? [2]

Before drilling starts at Site II, the engine of the MTZ borewell drilling machine breaks down.

- (iv) To save costs, the company considers to buy a new engine from an alternative manufacturer. The manufacturer claims that with the new engine, the MTZ borewell drilling machine will be able to drill a hole to a theoretical maximum depth of at least 360 m if it is operated continuously. Find the range of values of r for which the alternative manufacturer's claim is false. [3]

- (v) The company changes its mind and decides to buy a STK borewell drilling machine to drill the hole at Site II instead. Suppose that the machine performs the same way at Site II as at the test site, explain clearly whether the STK borewell drilling machine will be able to drill a hole of depth 150 m if it is operated continuously. [3]

STAPLE HERE

STAPLE HERE

[Turn over

- 11 [It is given that a cone of radius r with height h and slant edge l has lateral surface area $\pi r l$ and volume $\frac{1}{3}\pi r^2 h$.]

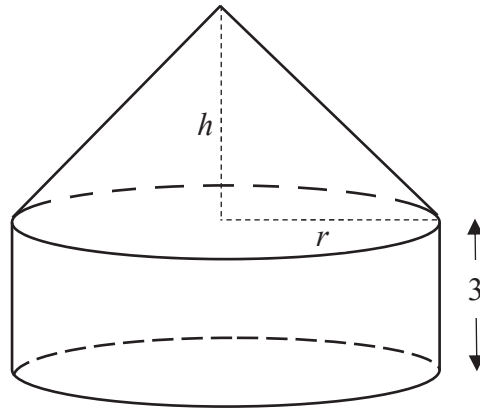


Figure 1

A circus tent consists of a cylindrical shape of radius r metres and height 3 metres and a right conical roof of height h metres as shown in Figure 1.

The conical roof has a fixed volume, V , and a lateral surface area, S , which varies with its height and radius.

- (i) Show that $S^2 = \frac{9V^2 + \pi^2 r^6}{r^2}$. [3]

- (ii) Using differentiation, find the ratio $h : r$ where the least amount of material is used for the conical roof . Justify that this ratio corresponds to the least amount of material used. [5]

STAPLE HERE

STAPLE HERE

[Turn over

The circus tent is firmly anchored to a horizontal ground using taut ropes as shown in Figure 2. Each rope is inclined at an angle of $\frac{\pi}{6}$ with the ground and makes an angle of $\frac{\pi}{3}$ with the perpendicular height of the cylindrical shape of the tent.

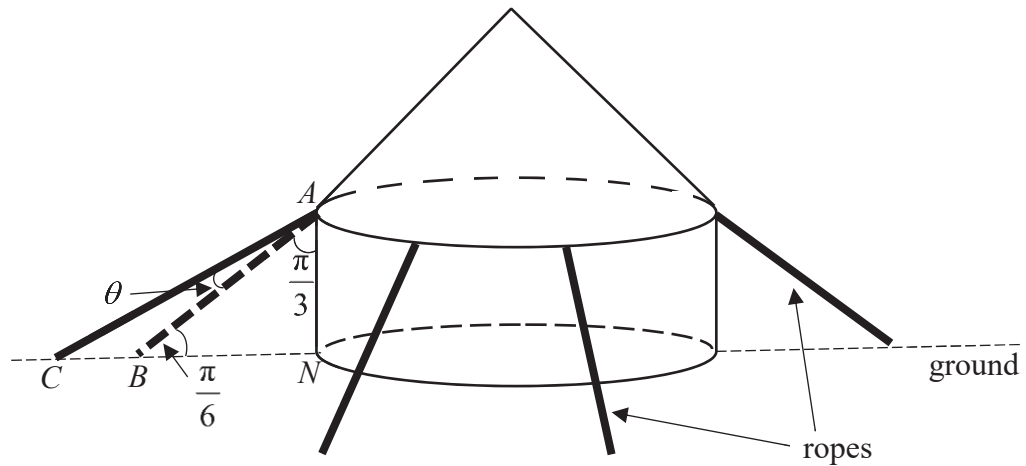


Figure 2

When anchoring a rope to the ground from point A to point B , an error of θ radians was made resulting in the rope being anchored on the ground at point C instead. Point N lies directly below point A such that points C , B and N are collinear.

- (iii) Show that the length of the rope AC is $\frac{3}{\sin\left(\frac{\pi}{6} - \theta\right)}$ metres. [2]

- (iv) To rectify the error, part of the rope AC is to be cut off so that the remaining part can be anchored tautly on the ground at point B .

Given that θ is a sufficient small angle, deduce an approximation for the length of the rope to be cut off, in ascending powers of θ up to and including the term in θ^2 . [3]